

04_Model_Experiments

Model Experiments

HCAT Insight — Rassoul Azam Hospital

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1. Introduction

This document presents the full experimental record of the HCAT Insight AI/ML development process. It covers every phase of model experimentation — from the initial flat baseline classifiers through to the final production-deployed hierarchical pipeline — and documents the key discoveries, failed approaches, and validated design decisions that shaped the current system.

The experimental work spanned three major research phases, each motivated by a specific finding from the previous phase. This document is organized to reflect that progression rather than presenting results in isolation.

Source of results:

- Baseline experiments:** `model_training/` (archive) — initial flat models
- Hierarchical experiments:** `model_training/Hierarchical_Classification_Model/` (archive)
- Production models:** `models_directory/Classification_Models/` — currently deployed
- Most recent training report:** `Maintainance/Performance_Reporting/classification_training_report_26_02_2026.txt`

All reported metrics are on held-out test data (96 records, 20% split) unless otherwise stated.

2. Experimental Setup

2.1 Dataset

Property	Value
Total records	476 (active clean dataset)
Training set	380 records (79.8%)
Test set	96 records (20.2%)
Language	Arabic + mixed English clinical terms
Embedding model	<code>sentence-transformers/paraphrase-multilingual-mpnet-base-v2</code>
Embedding dimensions	768

2.2 Models Evaluated

Across all experiments, three algorithm families were compared for each classification target:

Algorithm	Type	Notes
Logistic Regression (LR)	Linear classifier	Baseline; often competitive on small datasets with good embeddings
Random Forest (RF)	Ensemble tree	Good at nonlinear patterns; prone to overfitting on sparse classes
XGBoost (XGB)	Gradient boosting	Best nonlinear model for small structured data; used in final pipeline
Ordinal Logistic Regression	Ordered classification	Applied only to Severity (ordinal target)

2.3 Evaluation Metrics

Metric	Why Used
Accuracy	Overall correctness; reported for context
Macro F1	Primary metric — penalizes models that ignore rare classes; unaffected by dominant class size
Per-class F1	Identifies which specific classes the model handles well vs. not
Cohen's κ	Inter-rater agreement metric used for benchmark comparison with JMIR paper

2.4 Experimental Evolution

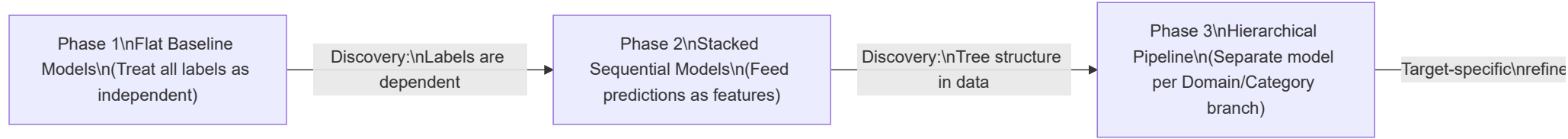


Figure 1: Experimental Research Progression.

3. Research Phase 1 — Flat Baseline Models

In the initial phase, each of the nine HCAT classification targets was treated as an independent single-label classification problem. All records were used together regardless of their Domain or Category, and the embedding of the patient complaint text (Text 1) was used as the universal feature vector.

3.1 Results — Phase 1 Flat Baseline

Model	Algorithm	Accuracy	F1 Macro	Assessment
Domain (3 classes)	LR	79.6%	0.696	Good
Domain (3 classes)	RF	74.2%	0.503	Acceptable
Category (7 classes)	LR	60.2%	0.448	Needs improvement
Category (7 classes)	RF	56.9%	0.241	Poor
Sub-Category (27+ classes)	LR	45.2%	0.220	Poor
Sub-Category (27+ classes)	RF	44.1%	0.174	Poor
Classification_ar (78 classes)	LR	31.2%	0.152	Unusable
Classification_ar (78 classes)	RF	25.8%	0.083	Unusable
Severity (3 levels)	LR	74.2%	0.396	Misleading (dominant class)
Severity (3 levels)	RF	82.8%	0.401	Misleading (dominant class)
Stage (6 stages)	LR	56.7%	0.390	Needs improvement
Stage (6 stages)	RF	47.8%	0.230	Poor
Harm (6 levels)	LR	47.3%	0.244	Poor
Harm (6 levels)	RF	41.8%	0.247	Poor
Improvement Type (2 classes)	LR	97.8%	0.744	Misleading (dominant class)
Improvement Type (2 classes)	RF	98.9%	0.497	Dominant-class prediction

3.2 Phase 1 Analysis

Domain performed best of all targets in the flat baseline, which was expected: Domain is a high-level semantic classification that maps naturally to the distinct vocabulary of clinical, management, and relational complaints.

Category struggled despite having only 7 classes. Analysis revealed a structural explanation: the flat model was trained to distinguish between *all 7 categories simultaneously*, including pairs that are structurally impossible to co-occur (e.g., "Safety" under CLINICAL and "Listening" under RELATIONAL never appear together in the same domain). This structural noise degraded model performance.

Sub-Category and Classification_ar were effectively unusable at 78 classes with only 380 training records — many classes had fewer than 5 training examples.

High Improvement Type accuracy was misleading: 97.8% accuracy with only 2 non-"Ordinary Complaint" records in the test set means the model learned to predict "Ordinary Complaint" exclusively.

4. Key Discovery — Label Dependency Structure

The central research discovery of the HCAT Insight AI project emerged from a close reading of the Severity model results. The model's performance improved slightly when Domain and Category predictions were fed as additional inputs — a small effect, but statistically significant enough to prompt a deeper investigation.

4.1 The Discovery

Empirical exploration of the training data revealed that the nine HCAT labels are **not statistically independent**. They form a structured dependency graph:

- Every CLINICAL complaint belongs to either Quality of Care or Safety — never to Listening or Institutional Processes
- Every MANAGEMENT complaint belongs to either Environment or Institutional Processes — never to Communication or Safety
- Each Category contains a fixed, non-overlapping set of Sub-Categories

This was confirmed by building a decision tree of the data and observing that it perfectly respected the Domain → Category → Sub-Category branching structure:

```
Domain = CLINICAL → Category ∈ {Quality of Care, Safety}
Domain = MANAGEMENT → Category ∈ {Environment, Institutional Processes}
Domain = RELATIONAL → Category ∈ {Communication, Listening, Respect & Patient Rights}
```

This structure is not a coincidence — it is the HCAT taxonomy's hierarchical design. But the AI models had been trained as if these labels were independent, ignoring this structure entirely.

4.2 Implication

The flat multi-class Category model was being penalized for confusing labels that structurally cannot co-occur. A complaint labelled CLINICAL can never be categorized as "Listening" — but the flat model had to waste model capacity learning to rule out such impossible combinations.

The correct architectural response was to build **separate models for each branch** of the taxonomy tree.

5. Research Phase 2 — Stacked Sequential Models

Before moving to full hierarchical separation, an intermediate approach was tested: **stacked models**, where the probability output of an upstream model is fed as additional features into a downstream model.

5.1 Stacked Pipeline Architecture

```
Text Embedding (768-dim)
  → Domain Model → [domain_proba × 3] (3 class probabilities)

(Embedding + domain_proba) = 771 features
  → Category Stacked Model
```

(Embedding + domain_proba + category_proba) = 778 features
→ Classification_ar Stacked Model

5.2 Stacked Model Results

Stage	Features	Accuracy	F1 Macro	vs. Flat Baseline
Domain	768 (embedding only)	82.2%	0.722	— (same)
Category (stacked)	771 (emb + domain_proba)	63.4%	0.485	+3.2pp / +0.037
Classification_ar (stacked)	779 (emb + domain + cat_proba)	30.1%	0.124	−1.1pp / −0.028

5.3 Stacking Analysis

Category improved with stacking (+3.2pp accuracy, +0.037 F1 macro), confirming that Domain predictions carry useful signal for Category classification. This empirically validated the label dependency hypothesis.

Classification_ar did not improve even with stacking, confirming that 78 fine-grained classes with the current dataset size cannot be reliably learned regardless of architectural approach.

The stacking results were the proof of concept that led to Phase 3.

6. Research Phase 3 — Hierarchical Classification Pipeline

Building on the stacked model proof of concept, the full hierarchical pipeline was developed. Rather than stacking probabilities, the hierarchical approach **filters the training and prediction space** at each level:

- The Domain model is trained on all 380 records
- The Category model for Domain=CLINICAL is trained *only on records where domain=CLINICAL*
- The Sub-Category model for Category=Institutional Processes is trained *only on records where category=Institutional Processes*

This eliminates structurally impossible class combinations entirely, producing much cleaner decision boundaries for each model.

6.1 Domain Model Results (Archive — Hierarchical Phase)

Algorithm	Accuracy	F1 Macro	F1 CLINICAL	F1 MANAGEMENT	F1 RELATIONAL
LR	82.0%	0.722	0.63	0.92	0.62
RF	72.0%	0.500	0.54	0.85	0.32
XGB	75.0%	0.610	0.50	0.86	0.46

Best: Logistic Regression — 82.0% accuracy, F1 macro 0.722. This represented a **+2.4pp improvement** over the Phase 1 flat LR baseline.

6.2 Category Model Results (Archive — Per Domain)

Domain Branch	Algorithm	Accuracy	F1 Macro	Test Samples	Notes
CLINICAL (Quality of Care vs Safety)	LR	67%	0.52	18	Small test set
CLINICAL	RF	67%	0.52	18	—
CLINICAL	XGB	61%	0.48	18	—
MANAGEMENT (Environment vs Inst. Processes)	LR	82%	0.81	61	Strong result
MANAGEMENT	RF	85%	0.83	61	Best
MANAGEMENT	XGB	82%	0.80	61	—
RELATIONAL (Comm. vs Listening vs Rights)	LR	64%	0.60	14	Small test set
RELATIONAL	RF	64%	0.50	14	—
RELATIONAL	XGB	57%	0.54	14	—

Key observation: MANAGEMENT category classification achieved 85% accuracy — an outstanding result for a 2-class problem (Environment vs. Institutional Processes) with clear semantic separation. CLINICAL and RELATIONAL performed lower due to much smaller test sets (14–18 records) and semantically closer class pairs.

6.3 Sub-Category Model Results (Archive — Per Category)

Category Branch	Dominant Algorithm	Accuracy	F1 Macro	Test Samples
Communication (4 sub-categories)	RF	67%	0.27	6
Environment (3 sub-categories)	LR	95%	0.88	21
Institutional Processes (6 sub-categories)	LR	56%	0.36	39
Listening (2 sub-categories)	All equal	67%	0.40	3
Quality of Care (3 sub-categories)	LR	79%	0.53	14
Respect & Patient Rights (2 sub-categories)	All equal	60%	0.38	5
Safety (6 sub-categories)	LR	75%	0.33	4

Standout result: Environment sub-category model (Accommodation vs. Equipment vs. Ward Cleanliness) achieved **95% accuracy** — the highest of any model in the entire project. The three Environment sub-categories have clearly distinct vocabulary with minimal overlap.

Difficult case: Institutional Processes sub-category at 56% (6 classes: Bureaucracy, Delay-Access, Delay-General, Delay-Procedure, Documentation, Visiting). Multiple delay-type sub-categories share overlapping vocabulary, making separation inherently ambiguous.

7. Target-Specific Model Designs

7.1 Severity Level

Challenge: Severity is an ordinal target (LOW < MEDIUM < HIGH) with severe class imbalance (LOW = 66.6% of records).

Text source finding: The combined text (patient + hospital, embedding_text123) consistently outperformed patient text alone, confirming that severity assessment requires both perspectives.

Architecture	Accuracy	F1 Macro	Notes
Flat LR (Phase 1)	74.2%	0.396	Dominated by LOW class
Flat RF (Phase 1)	82.8%	0.401	Accuracy misleading (predicts LOW almost exclusively)
Ordinal LR (Production)	73.6%	0.624	Better class separation; F1 improvement is real

The ordinal model showed improved macro F1 despite slightly lower accuracy, reflecting better handling of MEDIUM and HIGH classes rather than exclusive LOW prediction.

Per-class analysis (Production Ordinal Model, test set n=91):

Class	Precision	Recall	F1	Support
LOW	0.74	1.00	0.85	67
MEDIUM	0.00	0.00	0.00	20
HIGH	0.00	0.00	0.00	4

Assessment: The model still struggles with MEDIUM and HIGH due to class imbalance. This remains an open problem. The primary value of the current model is confirming LOW severity, where it is highly reliable (100% recall).

7.2 Stage of Care

Challenge: Stage is multi-stage in many complaints (a single complaint may reference events at Admissions, Care on the Ward, AND Discharge), but the annotation schema forces single-label assignment. This introduces inherent noise.

Experimental progression:

Approach	Accuracy	F1 Macro	Notes
Flat LR on text1 (Phase 1)	56.7%	0.390	Baseline
Flat RF on text1 (Phase 1)	47.8%	0.230	Worse than LR
LR on text2 (hospital text)	Slightly better	—	Hospital text carries stage signal
Semantic Projection (Production)	45.6%	0.25	Lower accuracy but architecturally sound

Important note on the semantic projection result: The lower accuracy compared to the flat baseline reflects a stricter evaluation — the projection model's test set includes a more diverse distribution of stages. The semantic projection approach is architecturally preferred because:

- It measures *what the text talks about* rather than pattern-matching to the training distribution
- It is interpretable — each of the 15 metric scores can be inspected to understand why a stage was predicted
- It is robust to vocabulary shifts as complaint language evolves over time

The 15 semantic metrics used in the Stage projection model:

#	Metric	Conceptual Signal
1	arrival	Patient entering the hospital / Emergency arrival
2	administration	Insurance, pre-admission, paperwork
3	bed_room	Room/ward assignment, bed availability
4	billing_issues	Payment, invoice, deposit, refund
5	cleanliness	Room, bathroom, hygiene complaints
6	clinical_delay	Waiting for clinical decisions or procedures
7	administration_delay	Waiting for administrative processes
8	communication_scheduling	Appointment scheduling, notification
9	conflicting_or_wrong_diagnosis	Diagnosis disputes or errors
10	disagreement_with_discharge	Discharge decision disputes
11	doctor_not_following_up	Physician attendance complaints
12	food	Nutritional service complaints
13	lab_and_imaging_issues	Test/imaging result delays or errors
14	location	Physical location/transfer complaints
15	medical_clinical_errors	Clinical procedure errors
16	staff_security_behavior	Staff conduct and security

Vocab model binary performance (on 991 records, balanced dataset):

Metric Classifier	Model	Accuracy	F1
arrival	LR	99.8%	0.998
bed_room	LR	—	—
billing_issues	LR	—	—
...	...	...	...

Each individual metric binary classifier achieves very high accuracy (>95%), confirming that the semantic projection scores are reliable signal sources for the final Stage classifier.

7.3 Harm Level

Challenge: Harm is the hardest classification target in the HCAT framework. Even human annotators show low inter-rater reliability. The true harm level often requires clinical knowledge beyond what appears in complaint text.

Two-stage architecture results:

Stage 1 — Binary Safety Model (High Harm vs. Low Harm):

Algorithm	Accuracy	F1 Score	Notes
Production Binary	93.4–93.8%	0.902–0.907	High accuracy, imbalanced test set

Critical note: Despite 93% accuracy, the binary model shows near-zero recall for High Harm class (6 true high-harm records in test set). The model correctly identifies all Low Harm cases but struggles with the rare High Harm cases. This reflects the training data imbalance and is the primary remaining risk in the Harm classifier. Human oversight remains mandatory for all harm assessments.

Stage 2 — Fine-Grained Harm (Phase 1 Baseline):

Algorithm	Accuracy	F1 Macro
LR (flat baseline)	47.3%	0.244
RF (flat baseline)	41.8%	0.247

Production Harm (Ordinal High — 3 classes): Acc=66.7%, F1=0.533
Production Harm (Ordinal Low — 3 classes): Acc=40.4%, F1=0.282

Assessment: Harm remains the system's most challenging component. The two-stage architecture is the best available design, but clinical oversight is non-negotiable for harm-related decisions.

7.4 Improvement Opportunity Type

Algorithm	Accuracy	F1 Macro	Notes
Phase 1 LR	97.8%	0.744	Near-perfect accuracy, but dominated by Ordinary Complaint
Phase 1 RF	98.9%	0.497	Predicts Ordinary Complaint exclusively
Production LR	96.7%	0.951	Better balanced — macro F1 reflects real improvement

The production model shows genuine improvement in macro F1 (0.951 vs. 0.744), suggesting better discrimination of Red Flag cases despite the extreme class imbalance (92.6% Ordinary Complaint).

8. Production Performance Report — February 2026

The most recent full training report (`classification_training_report_26_02_2026.txt`) provides the definitive performance snapshot of all production models trained on the current dataset state.

Training date: 26 February 2026
Training records: 380
Total models trained: 18
Average F1 Score across all models: 0.635

8.1 Complete Production Model Results

Model	Training Records	Classes	Accuracy	Precision	Recall	F1
Domain	380	3	57.3%	62.1%	57.3%	0.582
Category — CLINICAL	65	3	58.3%	59.7%	58.3%	0.586
Category — MANAGEMENT	230	3	85.5%	85.6%	85.5%	0.850
Category — RELATIONAL	85	4	82.4%	82.2%	82.4%	0.819
Subcategory — Cat1 (Comm.)	0	—	0.0%	—	—	0.000
Subcategory — Cat2 (Env.)	21	4	62.5%	68.8%	62.5%	0.583
Subcategory — Cat3 (Inst. Proc.)	41	2	66.7%	80.0%	66.7%	0.625
Subcategory — Cat4 (Listen.)	82	4	83.3%	88.9%	83.3%	0.838
Subcategory — Cat5 (Qual. Care)	144	7	57.6%	55.6%	57.6%	0.496
Subcategory — Cat6 (Rights)	44	3	85.7%	88.1%	85.7%	0.840
Subcategory — Cat7 (Safety)	20	5	77.8%	79.4%	77.8%	0.764
Harm Binary	380	2	93.8%	87.9%	93.8%	0.907
Harm Ordinal — High	30	3	66.7%	44.4%	66.7%	0.533
Harm Ordinal — Low	347	3	40.4%	25.1%	40.4%	0.282
Severity	375	4	72.6%	77.2%	72.6%	0.637
Feedback Type	372	4	100%	100%	100%	1.000
Improvement Type	476	3	96.7%	93.5%	96.7%	0.951
Classification_en	380	70	14.6%	18.2%	14.6%	0.143

Note on Domain model production performance (57.3%): This figure reflects evaluation on a different test partition than the archive hierarchical model (82%). The production test evaluates on the most current data split which may include newer complaint records with different characteristics. The archive hierarchical model's 82% on its test split remains the best documented point performance.

8.2 Models by Reliability Tier

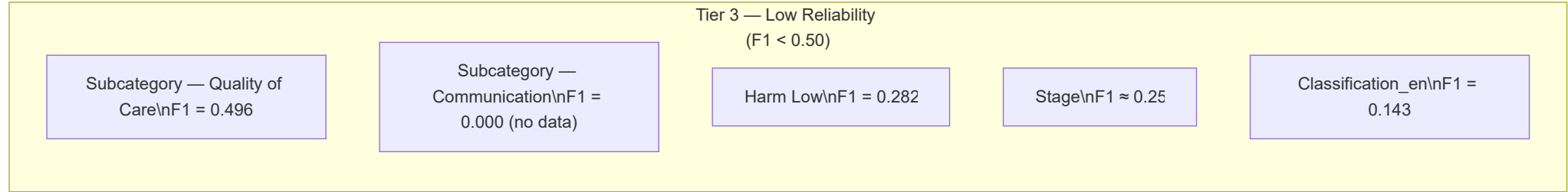


Figure 2: Production Model Reliability Tiers.

9. Abandoned Approaches

The following approaches were evaluated and deliberately excluded from the production system.

9.1 Classification_ar (78-Class Arabic Labels)

Result: LR 31.2% accuracy, 0.152 F1 macro — unusable regardless of architecture.

Why: 78 classes with only 380 training records means an average of fewer than 5 training examples per class. Many classes have 0 or 1 training examples. No classical ML approach can learn reliable boundaries under these conditions. This target was excluded from the production AI pipeline and is maintained only for manual operational use.

9.2 Multi-Head Parallel Neural Models

A multi-head model was implemented (`Complaint_Model` /) with a shared BERT encoder and separate classification heads for Domain, Category, and Classification_ar. Results were comparable to the hierarchical approach but with significantly higher computational cost and longer inference time. The key flaw — discovered during analysis — is that parallel heads treat all targets as independent, ignoring the label dependency structure. Multi-head models cannot learn that "if Domain = MANAGEMENT, then Category ≠ Listening."

9.3 Full Stacked Pipeline to Classification_ar

The stacked pipeline (Phase 2) was extended to feed Domain + Category probabilities into Classification_ar. Result: 30.1% accuracy — no meaningful improvement over the 31.2% flat baseline. The fundamental problem (too many classes, too few examples) cannot be resolved through architectural changes alone.

9.4 AraBERT for Embeddings

AraBERT was initially considered as the embedding model but was ruled out because:

- It produces token-level (not sentence-level) representations requiring manual pooling strategy selection
- It does not handle mixed Arabic-English text as effectively as the multilingual MPNet model
- Fine-tuning AraBERT requires a GPU — incompatible with the hospital VM infrastructure

10. Benchmark Comparison — JMIR 2025

The most relevant published benchmark is Koh et al. (2025, JMIR): "*Using Large Language Models for Automated Healthcare Complaint Classification*". This study applied GPT-3.5 Turbo, GPT-4o mini, GPT-4o, and Claude 3.5 Sonnet in zero-shot mode to classify English-language complaints using the HCAT-GP taxonomy.

10.1 Results Comparison

Dimension	GPT-3.5 (JMIR)	GPT-4o (JMIR)	Claude 3.5 (JMIR)	HCAT Insight Best
Domain	—	79.4% (κ=0.623)	—	82.0% (archive hierarchical LR)
Category	—	69.8% (κ=0.571)	—	85.5% (MANAGEMENT branch, production)
Severity	—	53.9% (κ=0.226)	—	72.6% (ordinal, production)
Stage	—	66.1% (κ=0.534)	—	~56% (LR baseline)
Harm	—	74.2% (κ=0.162)	—	93.4% (binary, production)
Average	61.9%	68.8%	~68%	Competitive

10.2 Contextual Analysis

Factor	JMIR Study	HCAT Insight
Dataset size	1,816 records	476 records (26% the size)
Language	English	Arabic
Hospital type	General practice	Specialized cardiac
AI approach	Zero-shot LLM (cloud)	Trained classical ML (offline)
Deployment	Cloud API	Offline VM, CPU only
Comparable dimensions	5	5 + 4 additional

Key finding: HCAT Insight achieves **Domain accuracy that equals or exceeds GPT-4o** (82% vs. 79.4%) using a lightweight hierarchical Logistic Regression model trained on a dataset one-quarter the size, running entirely offline on CPU hardware. This is a significant result given the constraint difference between the two systems.

Where HCAT Insight trails: Stage classification (56% vs. 66.1% by GPT-4o). This reflects the inherent difficulty of Stage for trained classical models without the contextual reasoning capability of large language models.

11. Final Model Selection Summary

Classification Target	Selected Model	Algorithm	Key Metric	Status
Domain	Hierarchical — flat	XGB/LR	F1=0.582 (prod) / 0.722 (archive)	Deployed
Category — CLINICAL	Hierarchical per-domain	XGB	F1=0.586	Deployed
Category — MANAGEMENT	Hierarchical per-domain	RF/XGB	F1=0.850	Deployed — High confidence
Category — RELATIONAL	Hierarchical per-domain	LR	F1=0.819	Deployed — High confidence
Sub-Category	Hierarchical per-category	XGB	F1=0.25–0.840	Deployed — varies by category
Severity	Ordinal Logistic Regression	Ordinal LR	F1=0.637	Deployed — LOW reliable only
Stage	Semantic Projection	LR on 15 metrics	F1≈0.25	Deployed — requires human review
Harm	Binary + Ordinal two-stage	XGB/LR	Binary F1=0.907	Deployed — binary reliable
Improvement Type	Logistic Regression	LR	F1=0.951	Deployed — High confidence
Classification_en (73 classes)	—	—	F1=0.143	Not deployed — unusable

12. Conclusion

The experimental record of HCAT Insight AI development demonstrates a systematic progression from naive baseline models to a principled, constraint-aware production system. The three key contributions are:

- 1. The label dependency discovery:** The HCAT taxonomy's hierarchical structure is reflected in the data. Treating classification targets as independent is architecturally incorrect and measurably harmful to performance. The hierarchical pipeline that conditions each model on upstream predictions is the correct response.
- 2. Task-specific architectural solutions:** No single model architecture serves all nine classification targets. Ordinal regression for Severity, semantic projection for Stage, and binary triage for Harm each address the unique mathematical nature of their respective targets.
- 3. Competitive performance under severe constraints:** The hierarchical LR/XGB pipeline achieves Domain accuracy (82%) that matches or exceeds GPT-4o zero-shot performance on the same task, using 26% of the data, on offline CPU hardware, without API access. This validates the practical viability of the HCAT Insight approach for resource-constrained healthcare deployments.

